ASSIGNMENT: 6

Introduction to Network Devices

Theory:

Task (1)

Device	Main Function	Typical Ports / Protocols
Switch	Connects multiple wired devices in the same local network and forwards data to the correct device.	Ethernet **RJ-45 ports**; commonly uses Ethernet (no specific TCP/UDP port)
Router	Connects the home LAN to the Internet and routes packets between networks.	**WAN/LAN Ethernet ports**; commonly uses **80 (HTTP), 443 (HTTPS), 53 (DNS), 67/68 (DHCP)**
Hub	Connects multiple wired devices and broadcasts incoming data to all ports.	Ethernet **RJ-45 ports**; no specific TCP/UDP port
Firewall	Filters incoming and outgoing traffic and blocks unauthorized connections.	Uses rules for ports such as **80, 443, 53, 22**, etc.
Access Point (AP)	Provides Wi-Fi connectivity so wireless devices can join the local network.	**Wi-Fi (802.11)**; Ethernet uplink; management may use **80/443**

Task (2):

Router (back panel):

- ❖ **Ethernet WAN Port (Blue):**
  - RJ-45 WAN (Internet) port used to connect an Ethernet cable to your broadband modem or ISP terminal.
- ❖ **Ethernet LAN Ports 1–4 (Yellow):**
  - Four RJ-45 LAN (Local Area Network) ports used for wired connections to local devices (e.g., laptop, desktop, switch, game console).

❖ **WPS / Wi-Fi Button:**

- Small push button used for Wi-Fi Protected Setup quick pairing and toggling Wi-Fi functionality.

Switch (front/back panel):

- Multiple identical-looking Ethernet ports (numbered) - all function the same way, connecting wired devices
- **Power port**
- (On managed switches only) a small console port - used for direct configuration via cable, not for regular data

Task (3):

❖ Example: Using Zomato

- When I use Zomato, the access point connects my phone to the Wi-Fi network.
- The router sends my request from the home network to the Internet and Zomato's servers.
- Firewalls help protect the network by blocking unauthorized traffic.
- Switches are used in data centers to connect servers and forward data efficiently.

Task (4):

Diagnosing slow/disconnecting college Wi-Fi:

- The Wi-Fi may be slow because **too many students are connected to the same access point**.
- The access point may not be able to handle so many devices and high traffic.
- Upgrading to a better, high-capacity access point can support more users.
- This can make the college Wi-Fi faster and more stable.